

- 13** An ideal gas undergoes an expansion in volume from  $1.3 \times 10^{-4} \text{ m}^3$  to  $3.6 \times 10^{-4} \text{ m}^3$  at a constant pressure of  $1.3 \times 10^5 \text{ Pa}$ . During this expansion, 24 J of heat is supplied to gas.

What is the overall change in internal energy of the gas?

- A** decrease of 54 J
- B** decrease of 6 J
- C** increase of 6 J
- D** increase of 54 J

- 14** The diagram shows the variation with distance  $d$  along a sinusoidal wave of displacement